

JEE MAIN REVISION MASTER: PERMUTATIONS AND COMBINATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Permutations, Combinations & Combinatorial Geometry	14 PYQs	Combinatorics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Permutations, Combinations & Combinatorial Geometry	${}^n P_r = \frac{n!}{(n-r)!}, \quad {}^n C_r + {}^n C_{r-1} = {}^{n+1} C_r, \quad \text{Diagonals} = \frac{n(n-3)}{2}, \quad \text{Triangles} = {}^n C_3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Permutations, Combinations & Combinatorial Geometry (High Yield — 14 PYQs)

Core Principle: Permutations (arrangements where order matters) ${}^n P_r = \frac{n!}{(n-r)!}$. Combinations (selections where order does not matter) ${}^n C_r = \frac{n!}{r!(n-r)!}$. Pascal's identity: ${}^n C_r + {}^n C_{r-1} = {}^{n+1} C_r$. Circular permutations of n distinct objects: $(n-1)!$ (for necklaces/garlands where clockwise and anticlockwise are identical: $\frac{(n-1)!}{2}$). Combinatorial geometry: Number of diagonals in an n -sided polygon = ${}^n C_2 - n = \frac{n(n-3)}{2}$.

■ **Caution / JEE Pitfall:** For necklaces or garlands of n beads, the number of arrangements is $\frac{(n-1)!}{2}$, NOT $(n-1)!$, because flipping over reverses direction.

■ **Speed Trick:** Number of handshakes among n people: ${}^n C_2 = \frac{n(n-1)}{2}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "10 distinct points are taken on a circle. Then using these points Statement I: The number of triangles that can be formed is 100 Stateme..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The number of four digit even number that can be formed using the digits 0,1,2 and 3 without repetition is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?

- (A) 720
- (B) 360
- (C) 120
- (D) 180

Correct Answer: None

Solution Analysis: The characters in "KCET 2025" are: K,C,E,T,2,0,2,5 There are 8 characters in total, and the digit 2 is repeated twice. We need arrangements that start with K and end with 5 . So fix K in the first position and 5 in the last position: K_____5 Now we have to arrange the remaining 6 characters: C,E,T,2,0,2 If all were different, the number of arrangements would be: $6!$ But the digit 2 appears twice, so we divide by $2!$: $(6! / 2!) = (720 / 2) = 360$ So the number of ways is: 360 Correct option: Option B .